Balanced symbol equations

Aim

To write balanced symbol equations for reactions.

Equipment

- Resource sheet E6a
- scissors

What to do

1. Cut out the cards on the resource sheet.
2. Arrange cards for hydrogen and chloride side by side:

H **hydrogen**	**Cl** **chloride**

   To match the cards you need one hydrogen card and one chloride card. This represents the formula for hydrochloric acid, which is HCl.

   (Note: All acids have hydrogen as part of the formula. Sulphuric acid is hydrogen sulphate and nitric acid is hydrogen nitrate.)

3. Now arrange the cards to show the formula for magnesium hydroxide:

Mg **magnesium**	**OH** **hydroxide**
	OH **hydroxide**

   Note that two hydroxide cards are needed to match up to one magnesium card, so the formula for magnesium hydroxide is $Mg(OH)_2$. The set of brackets around OH shows that we have two lots of OH. If we wrote $MgOH_2$ it would look like one oxygen and two hydrogens.

4. Next, you are going to write a balanced symbol equation for the reaction between magnesium hydroxide and hydrochloric acid.

   Write the formulae of the reactants:

   $Mg(OH)_2$ + HCl →

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Balanced symbol equations
continued

5 Now rearrange the cards to make the formulae of the products:

Mg magnesium	Cl chloride
	Cl chloride

H hydrogen	OH hydroxide

H hydrogen	OH hydroxide

Note that you need an extra Cl because the formula for magnesium chloride is $MgCl_2$. To get this extra Cl you will need an extra molecule of hydrochloric acid. Using the extra H, two molecules of water, H_2O, can be formed.

6 Now use the cards to help you complete the symbol equation:

$$\mathbf{Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O}$$

Note how the number 2 has been placed in front of the formulae for hydrochloric acid and for water, to show that two molecules of each are in the equation.

Questions

Use the same method to write a balanced symbol equation for each of the following reactions.

1 Nitric acid + potassium hydroxide

2 Magnesium + hydrochloric acid

3 Copper oxide + sulphuric acid

4 Zinc carbonate + hydrochloric acid (Hint: Carbonate, CO_3, splits up in this reaction. It might be helpful to make two extra oxygen cards.)

5 Zinc + sulphuric acid

6 Nitric acid + magnesium hydroxide

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Balanced symbol equations
continued

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